AI505

Optimization

Optimization in Machine Learning

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Outline

1. Analysis of SG

Theoretical Analysis — Preliminaries

convergence properties and worst-case iteration complexity bounds.

$$F(\mathbf{w}) = \begin{cases} R_n(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{w}) & \text{Empirical Risk} \\ R(\mathbf{w}) = \mathbb{E}_{\xi}[f(\mathbf{w};\xi)] & \text{Expected Risk} \end{cases}$$

sampling uniformly with replacement from training set $\rightsquigarrow R_n$
sampling with $P(\xi)$ with replacement from training set $\rightsquigarrow R$.

Procedure SG(...);
Choose an initial iterate $\mathbf{w}_0$;
for $k = 0, 1, \ldots$ **do**
  Generate a realization of the random variable ξ_k;
  Compute a stochastic vector $g(\mathbf{w}_k, \xi_k)$;
  Choose a stepsize $\alpha_k > 0$;
  Set the new iterate as $\mathbf{w}_{k+1} \leftarrow \mathbf{w}_k - \alpha_k g(\mathbf{w}_k, \xi_k)$;

Theoretical Analysis — Preliminaries

ξ_k may represent a single sample or a mini-batch
g may represent a stochastic gradient (biased estimator of $\nabla F(\mathbf{w}_k)$ or a stochastic Netwon or quasi-Newton direction).

$$g(\mathbf{w}_k, \xi_k) = \begin{cases} \nabla f(\mathbf{w}_k; \xi_k) \\ \frac{1}{n_k} \sum_{i=1}^{n_k} \nabla f(\mathbf{w}_k; \xi_{k,i}) \\ H_k \frac{1}{n_k} \sum_{i=1}^{n_k} \nabla f(\mathbf{w}_k; \xi_{k,i}) \end{cases}$$

H_k a symmetric positive definite scaling matrix (we called it B_k^{-1} earlier in this course)
α_k fixed stepsize or diminishing stepsizes
$\mathbf{w}_k$ can have influence on the sample selection (active learning)

Convergence Analysis – Assumptions

- Assumption 4.1 Lipschitz-continuous objective gradients:
  $\|\nabla F(\mathbf{w}) - \nabla F(\bar{\mathbf{w}})\|_2 \leq L\|\mathbf{w} - \bar{\mathbf{w}}\|_2$ for all $\{\mathbf{w}, \bar{\mathbf{w}}\} \subseteq \mathbb{R}^d$
- Assumption 4.3 First and second moment limits. The objective function and Procedure SG satisfy the following:
  - objective function is bounded from below by a scalar F_{inf} over the region explored by the algorithm.
  - in expectation, the vector $-g(\mathbf{w}_k, \xi_k)$ is a direction of sufficient descent for F from $\mathbf{w}_k$ with a norm comparable to the norm of the gradient
  - the variance of $g(\mathbf{w}_k, \xi_k)$ is restricted, but in a relatively minor manner.

$$\mathrm{Var}_{\xi_k}[g(\mathbf{w}_k, \xi_k)] \leq M + M_V \|\nabla F(\mathbf{w}_k)\|_2^2, \; M > 0, M_V > 0 \text{ for all } k \in \mathbb{N}$$

- Lemma: SG is a Markovian process in the sense that $\mathbf{w}_{k+1}$ is a random variable that depends only on the iterate $\mathbf{w}_k$, the seed ξ_k, and the stepsize α_k and not on any past iterates.

Convergence Analysis – Assumptions

- Assumption 4.5 Strong convexity.
  The objective function $F : \mathbb{R}^d \to \mathbb{R}$ is **strongly convex** in that there exists a constant $c > 0$ such that

$$F(\bar{\mathbf{w}}) \geq F(\mathbf{w}) + \nabla F(\mathbf{w})^T(\bar{\mathbf{w}} - \mathbf{w}) + \frac{1}{2}c\,\|\bar{\mathbf{w}} - \mathbf{w}\|_2^2 \qquad \text{for all } (\bar{\mathbf{w}}, \mathbf{w}) \in \mathbb{R}^d \times \mathbb{R}^d$$

  or equivalently if there exists $c > 0$:

$$\nabla^2 F(\mathbf{w}) \succeq c$$

  (for univariate case: $f''(w) \geq c$), ie, grows at least quadratically.

  Hence, F has a unique minimizer, denoted as $\mathbf{w}^* \in \mathbb{R}^d$ with $F^* \stackrel{def}{=} F(w^*)$.

Convergence Analysis – Results

- Theorem 4.6 (Strongly Convex Objective, Fixed Stepsize).
  - optimality gap has an asymptotic bound in expectation depending on $\bar{\alpha}$.
  - sublinear convergence rate of the suboptimality gap
- Theorem 4.7 (Strongly Convex Objective, Diminishing Stepsizes)
  SG with diminishing step size converges in expectation.
  - role of strong convexity
  - role of initial point
  - trade-offs of mini batches
- Theorem 4.8 (Nonconvex Objective, Fixed Stepsize)
  - While one cannot bound the expected optimality gap as in the convex case, inequality (4.28b) bounds the average norm of the gradient of the objective function observed on $\{\mathbf{w}_k\}$ visited during the first K iterations.
  - classical result for the full gradient method applied to nonconvex functions, namely, that the sum of squared gradients remains finite, implying that

$$\{\|\nabla F(\mathbf{w}_k)\|_2\} \to 0.$$

- Theorem 4.9 (Nonconvex Objective, Diminishing Stepsizes)
  - for the SG method with diminishing stepsizes, the expected gradient norms cannot stay bounded away from zero
  - the weighted average norm of the squared gradients converges to zero even if the gradients are noisy, (i.e., if $M > 0$ in the Variance upper bounding assumption) one can still conclude that the expected gradient norms cannot asymptotically stay far from zero.

Computational Complexity Analysis

- consider a big data scenario with an infinite supply of training examples, but a **limited computational time budget**. what type of algorithm — e.g., a simple SG or batch gradient method — would provide the best guarantees in terms of achieving a low expected risk?

- $\mathbf{w}^* \in \text{argmin}\, R(\mathbf{w})$; $\mathbf{w}_n \in \text{argmin}\, R_n(\mathbf{w})$, $\tilde{\mathbf{w}}_n$ approximate empirical risk minimizer returned by a given optimization algorithm at $\mathcal{T}_{max}$

- The tradeoffs associated with this scenario can be formalized as choosing the family of prediction functions $\mathcal{H}$, the number of examples n, and the optimization accuracy $\epsilon \stackrel{def}{=} E[R_n(\tilde{\mathbf{w}}_n) - R_n(\mathbf{w}_n)]$ in order to minimize the total error:

$$\underset{\mathcal{H}, n \in \mathbb{N}, \epsilon}{\text{minimize}}\; E[R(\tilde{\mathbf{w}}_n)] = \overbrace{R(\mathbf{w}^*)}^{\mathcal{E}_{app}(\mathcal{H})} + \overbrace{E[R(\mathbf{w}_n) - R(\mathbf{w}^*)]}^{\mathcal{E}_{est}(\mathcal{H}, n)} + \overbrace{E[R(\tilde{\mathbf{w}}_n) - R(\mathbf{w}_n)]}^{\mathcal{E}_{opt}(\mathcal{H}, n, \epsilon)}$$

$$\text{subject to } \mathcal{T}(n, \epsilon) \leq \mathcal{T}_{max}$$

Computational Complexity Analysis

- SG, with its sublinear rate of convergence, is more efficient for large-scale learning than (full, batch) gradient-based methods that have a linear rate of convergence.

- reducing the optimization error $\mathcal{E}_{opt}(\mathcal{H}, n, \epsilon)$ (evaluated with respect to R rather than R_n) one might need to make up for the additional computing time by: (i) reducing the sample size n, potentially increasing the estimation error $\mathcal{E}_{est}(\mathcal{H}, n)$; or (ii) simplifying the function family $\mathcal{H}$, potentially increasing the approximation error $\mathcal{E}_{app}(\mathcal{H})$.

Computational Complexity Analysis

Instead, keep fixed $\mathcal{H}$ carrying out a worst-case analysis on the influence of the sample size n and optimization tolerance ϵ, which together only influence the estimation and optimization errors.

- this leads to an upper bound:

$$\mathcal{E}(n,\epsilon) = \mathcal{O}\left(\frac{\sqrt{\log n}}{n} + \epsilon\right)$$

- when the loss function is strongly convex

$$\mathcal{E}(n,\epsilon) = \mathcal{O}\left(\frac{\log n}{n} + \epsilon\right) \qquad \text{and when } n \text{ large to} \quad \mathcal{E}(n,\epsilon) \sim \frac{1}{n} + \epsilon$$

Computational Complexity Analysis

Recall:

- the total work required to obtain ϵ-optimality for SG is $\mathcal{T}_{stoch} \sim 1/\epsilon$,
- the total work required to obtain ϵ-optimality for a batch gradient method is $\mathcal{T}_{batch} \sim n \log(1/\epsilon)$.

	Batch	Stochastic
$\mathcal{T}(n, \epsilon)$	$\sim n \log\left(\frac{1}{\epsilon}\right)$	$\frac{1}{\epsilon}$
$\mathcal{E}^*$	$\sim \frac{\log(\mathcal{T}_{\max})}{\mathcal{T}_{\max}} + \frac{1}{\mathcal{T}_{\max}}$	$\frac{1}{\mathcal{T}_{\max}}$

A stochastic optimization algorithm performs better that batch stochastic in terms of expected error

Large gap between asymptotical behavior and practical realities.

Remarks

- Fragility of the Asymptotic Performance of SG
  ok if objective function includes a squared L_2-norm regularizer (related to constant c) but regularization parameter should be lowered when the number of samples increases.

- SG good for GPUs but ill-conditioning erodes efficiency of SG

- Distributed computing not working with basic SG because of too frequent updates of $\mathbf{w}$, more promising with mini-batch.

- Alternatives with Faster Convergence: minimizing empirical risk R_n there is information from previous gradients.
  - gradient aggregation methods
  - dynamic sampling approach